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Notifications



1. RLC circuit

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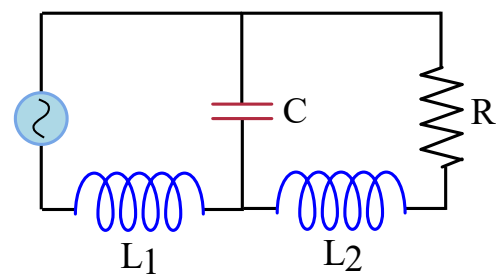
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In general, the voltage and current in any circuit with standard components can be modeled by a first-order system of ODE's. In the diagram above, suppose that the input (power supply) is given as $V_{in}(t) = \sin \omega t$. Let V_1 be the voltage drop across the inductor with inductance L_1 , and let I_1 be the current through that inductor. Similarly, let V_2 and I_2 be the voltage drop across and the current through the inductor with inductance L_2 . Let V_C be the voltage drop across the capacitor with capacitance C and let I_C be the current through the capacitor. Let V_R be the voltage drop across the resistor of resistance R , and let I_R be the current through the resistor.

There are eight dependent variables (I 's and V 's), and the eight equations governing the system are

- 1. $I_R = I_2$
- 2. $I_1 = I_C + I_2$
- 3. $V_C = V_R + V_2$
- 4. $V_C + V_1 = \sin \omega t$
- 5. $V_R = RI_R$
- 6. $\dot{V}_C = \frac{I_C}{C}$
- 7. $V_1 = L_1 \dot{I}_1$
- 8. $V_2 = L_2 \dot{I}_2$

RLC (a) (External resource) (1.0 points possible)

Find the matrix system

Let $\mathbf{x} = \begin{bmatrix} V_C \\ I_1 \\ I_2 \end{bmatrix}$. The equations above can be written as a 3×3 system

$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}(t)$.

Find $\mathbf{A}$ and $\mathbf{b}$.

Enter your answer in terms of the symbolic entries listed here and defined in the template below.

- R for R
- C for C
- L1 for L_1
- L2 for L_2
- w for ω
- t for t

```
1 % Create symbols used in defining A and b
2 R = sym('R');
3 C = sym('C');
4 L1 = sym('L1');
5 L2 = sym('L2');
6 w = sym('w');
7 t = sym('t');
8
9 %Define the matrix A and the vector b below.
10 %Note that b should be a column vector for the differential equation to make
11
12 A = [0, 1/C, -1/C; -1/L1, 0, 0; 1/L2, 0, -R/L2];
13 b = [0; sin(w*t)/L1; 0];
```

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Assessment: All Tests Passed

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-  Check the matrix A
-  Check the vector b

RLC (b)

1/1 point (graded)

Let $L_1 = L_2 = C = 1$. Find the characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A})$ (in terms of R).

(Enter as a polynomial in **lambda** with leading coefficient +1.)

lambda^3+R*lambda^2+2*lambda+R



Answer: lambda^3+R*lambda^2+2*lambda+R

$\lambda^3 + R \cdot \lambda^2 + 2 \cdot \lambda + R$

 FORMULA INPUT HELP

Solution:

The characteristic polynomial is $\det(\lambda \mathbf{I} - \mathbf{A})$, which is

λ

-1

1

1

λ

0

-1

0

$\lambda + R$

$= \lambda^2 (\lambda + R) + \lambda + (\lambda + R) = \lambda^3 + R\lambda^2 + 2\lambda + R.$

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You have used 1 of 3 attempts

 Answers are displayed within the problem

1. RLC circuit

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